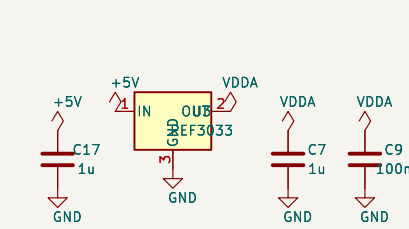
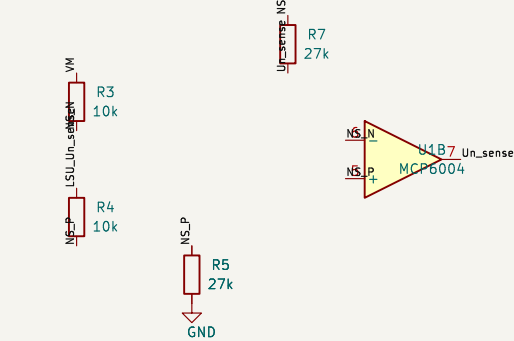


LSU 4.9 analog front-end - topology after rusEFI/FOME wideband module rev C (MIT). See FRONTEND.md

Analog reference VDDA = 3.3 V



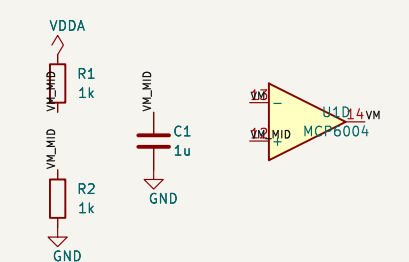
Nernst sense: $Un_{sense} = 2.7 * (Un - VM)$, target $Un - VM = 0.45 V$



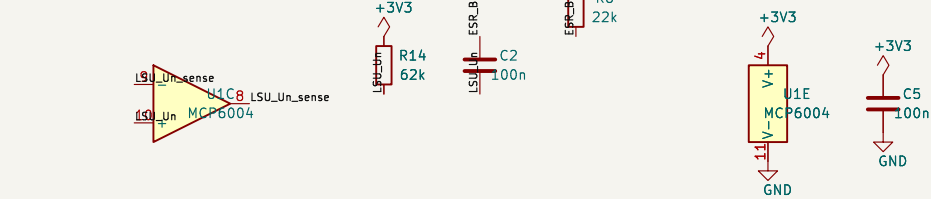
Power flags (ERC: nets fed from connectors / regulators)



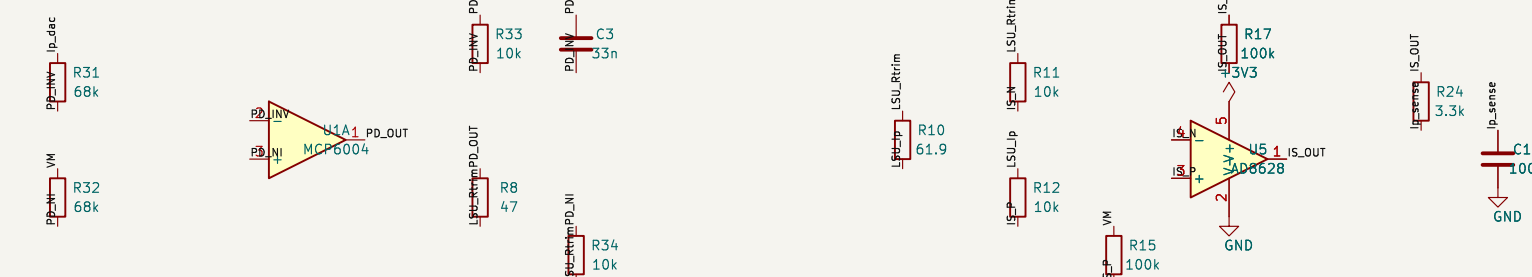
Virtual ground VM = VDDA/2



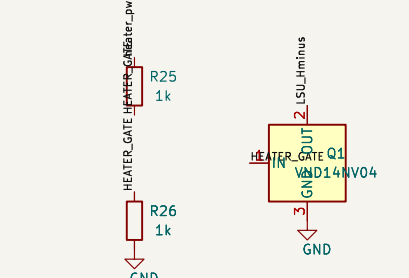
Nernst buffer, bias current (62k) and Ri injection (22k + 100n)



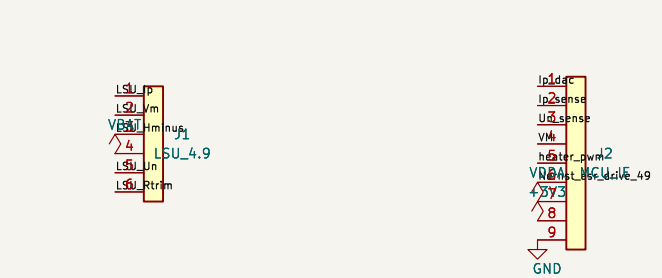
Pump driver: Howland current source, $I = -(R33/R31) * (Ip_{dac} - VM) / R_{pump}$ current sense: $Ip_{sense} = VM + 10 * (V_{Ip} - V_{Rtrim})$, shunt 61.9 ohm



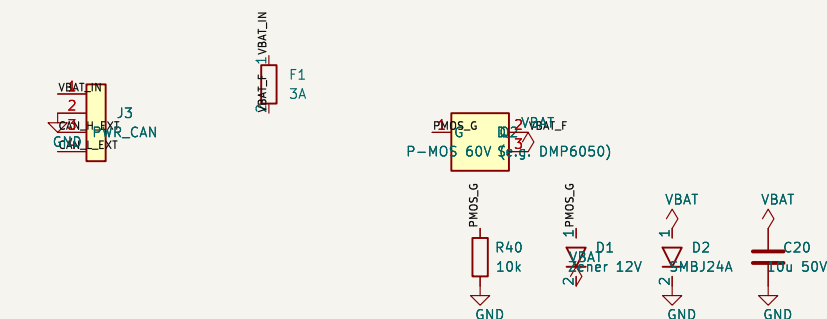
Heater switch



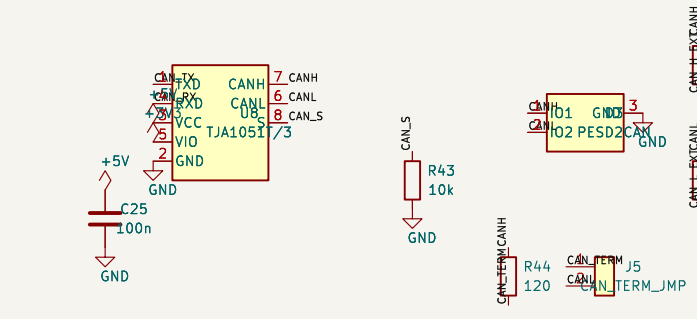
LSU 4.9 connector (pin order to be verified against MIT bench and datasheet)



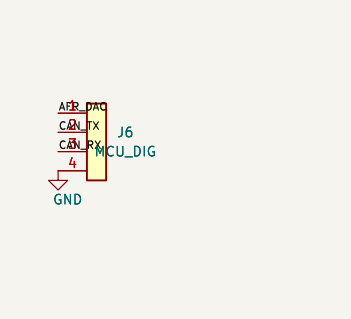
Power input: fuse, reverse-polarity P-MOSFET, TVS



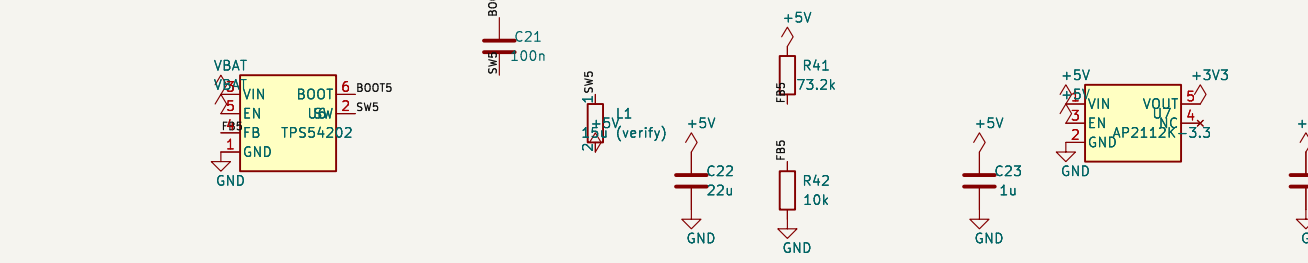
CAN (rusEFI wideband protocol)



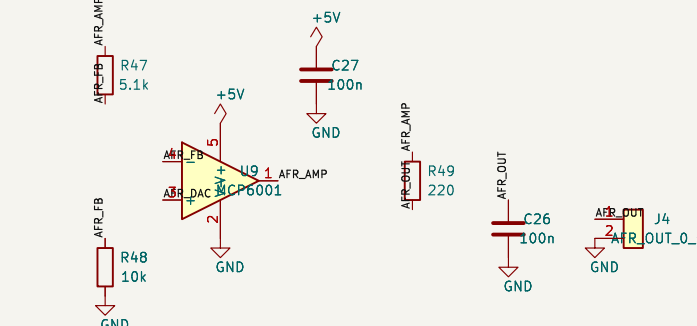
Interface to MCU board (digital)



5 V buck (TPS54202: pinout and values from memory - VERIFY against datasheet) 3.3 V LDO



0-5 V AFR output: 10.20 AFR = 0.5 V, DAC 0-3.3 V x1.51



Draft - NOT verified by ERC/simulation/bench
Topology after rusEFI/FOME wideband module rev C (MIT)